

Dixit Sabharwal

Software Engineer

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Profile

I am a full-stack developer, trying to stay ahead of the newest trends in building engaging and consistent frontends, as well as developing maintainable APIs that adhere to open standards.

Also trying to stay proficient in deploying these solutions across various environments; from Kubernetes clusters and serverless cloud functions to bare-metal VMs and on-premises servers.

I'm dedicated to learning and building systems that are reliable and practical but also scalable where needed. I am particularly interested in working on projects or with companies that support open-source software or work in the public sector. I'm convinced of the importance of making quality software accessible to everyone.

Experience

Software Engineer @ AWS Billing (Amazon Web Services)

Nov 2024 - Currently
Berlin, Germany

- Developing and operating a system of record storing the product catalog at AWS. The system operates in 6 regions (across US, EU, China) servicing over 90 million requests per day with over 99.95% availability. Operations include 24/7 on-call shifts, incident response and post mortems.
- I modernised our service's integration with the internal DataWarehouse to deliver a data freshness SLA of <10 minutes, by replacing snapshot exports that run every 8 hours with a CDC processing pipeline that wrote to an Apache Iceberg catalog. The Iceberg catalog also opened the doors to seamless integration with AWS and OSS tooling, reducing the barrier of entry for using this data for new tools.
- I also contributed to the storage-layer refactor of our service, where we migrated data from a MySQL cluster to DynamoDB tables, without any service downtime and while maintaining the 99.95% availability SLA of our apis.
- I have also driven our progress towards zero-touch environment creation, where we automate the process of spinning up a new production environment for our service (required for new regions launched by AWS). This included maximising infra-as-code for our compute and storage needs and automating the bootstrapping of configurations, databases, access-control, etc.
- I have contributed to the development of a web console through which the management of different AWS services configure and price their catalog and offerings. It is a functional interface that must enable the users to wrangle large amounts of data with integrations for a/b testing changes and built in approval processes for changes.
- I have gained a lot of experience with developing on the AWS serverless stack, consisting of "Lambda" functions, "ECS" containers, "StepFunction" state-machine orchestrators, "Glue" spark jobs, etc.

Software Developer @ Facultés IC, EPFL

June 2023 - Jul 2023
Lausanne, Switzerland

- Part of team that set-up an automatic grading system for programming assignments. Through a Moodle plugin and Kubernetes cluster the system allowed for easy scalability and fault-tolerance during peak times of the semester.
- Created a controller (using Spring and Java) that runs each submission as a Job on the Kubernetes cluster. The controller sets up the env for the grading task, and returns the outcomes to Moodle.
- Set-up CI/CD pipeline using GitLab CI that automatically deploys the controller and Moodle in a staging environment, for quick integration testing and feedback.

Data analyst @ Taskbase

Jul 2023 - Sep 2023
Zurich, Switzerland

- Built CLI tool which extracts data from CSV and Excel sources provided by clients and passes it through the Taskbase API, to speed up the onboarding process.
- Generated sample data for QA tasks, through prompt engineering and OpenAI GPT models, to compensate for the lack of real data from new clients.

Software Development Intern @ AWS/Amazon

Aug 2022 - Feb 2023
Berlin, Germany

- Built a management console to empower ongoing migration of service to a new data model and store.
- Console is a web-based UI to view and search through data, also compare between old and new versions, and trigger different tasks in the migration pipeline.
- Used React and TypeScript for the website UI.
Used the AWS serverless stack (Lambda, API Gateway, DynamoDB) for the backend.
Managed infrastructure in AWS CDK.

Teaching Assistant @ Faculty of EEMCS, TU Delft

Sep 2019 - Jun 2021
Delft, Netherlands

- Teaching Assistant for > 10 different courses for the Computer Science bachelors program.

Web Developer @ Dizconto Ltd.

Apr 2020 - Jul 2020
Den Haag, Netherlands

- Built a management console frontend for Dizconto, which provided social media marketing integrations.
- Planned website development, converted mockups into usable web presence with TypeScript, React and NodeJS.
- Also worked on implementing a development and deployment (CI/CD) pipeline using Terraform and AWS.

Education

M.Sc. in Computer Science

EPFL
Sep 2021 - Present

- Following courses on software security and privacy, compilers and type-systems, and large-scale data handling.

B.Sc. in Computer Science Engineering

TU Delft
Sep 2018 - July 2021

- Graduated cum laude with a GPA of 8.8.
- Thesis on "Formal verification of functional programs in Haskell and Agda".
- Minor in Quantum Science and Information.

Skills

Languages: Java/Kotlin, Golang,
Javascript/Typescript, Python, Scala/Haskell

UI: React.js, Remix, HTML/CSS

Server Frameworks: SpringBoot, Node.js/Express,
Gin

Data Analysis: Pandas, Scikit-learn

Database Technologies: SQLite, PostgreSQL,
MySQL, DynamoDB

Large Scale Data Wrangling: Apache Spark,
Hadoop

Deployment Environments: AWS, Azure,
Kubernetes (k8s), Docker

CI/CD: GitLab CI